MIL-C-22226A (OS)
22 February 1973
SUPERSEDING
MIL-C-22226 (NOrd)
9 November 1959

MILITARY SPECIFICATION

COPPER-NICKEL-MANGANESE ALLOY BARS, PLATES, AND SHEETS, LOW MAGNETIC EFFECT (FOR SPECIAL PURPOSE TOOLS AND EQUIPMENT)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers a 60-percent copper, 20-percent nickel, 20-percent manganese alloy (see table I) having low magnetic permeability and capable of developing high strengths by heat treatment. (See 6.1.)

1.2 Classification. The copper-nickel-manganese alloy covered by this specification shall be of one composition and shall be furnished in the following forms and tempers, as specified (see 6.2).

Form:
1. Bar
2. Plate
3. Sheet

Temper.
Condition R. Hot rolled
Condition A. Hot rolled and solution treated
Condition H. Hot rolled, solution treated, and cold worked.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 9530

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SPECIFICATIONS

Military

MIL-P-3993	Copper and Copper-Base Alloy Mill Products, Packaging of
MIL-M-19595	Magnetic Effect Limits for Non-Magnetic Equipment Used in the Proximity of Magnetic Influence Ordnance

STANDARDS

Federal

FED-STD-151	Metals, Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The raw materials used shall be such as to produce copper-base alloy bars, plates, and sheets in compliance with the requirements of this specification.

3.2 Chemical composition. The chemical composition of copper base alloy bars, plates, and sheets shall conform to the requirements shown in table I. The contractor shall furnish an analysis of each lot showing the percentage of each of the elements specified in table I.

3.3 Magnetic effect. Material covered by this specification shall comply with the requirements of MIL-M-19595.

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Table I

CHEMICAL COMPOSITION

Element	Weight percent
Copper	Remainder
Nickel	19 - 21
Manganese	19 - 21
Iron	0.15, max.
Other	0.15, max.

3.4 Dimensions and tolerances. Dimensions and tolerances of bars, plates, and sheets shall be as specified in the contract or order (see 6.2).

3.5 Workmanship. Bars, plates, and sheets shall be uniform in quality and condition, shall be clean, sound, and smooth, and shall be free from injurious defects including foreign material, pipes, slivers, laps, cracks, twists, scale, damaged ends, corners, or edges, which due to their nature, degree, or extent, detrimentally affect the suitability of the material for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of all material of the same composition, form, and size from the same melt, heat-treated under the same conditions, and submitted for inspection at one time.

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4.3 Sampling.

4.3.1 For visual and dimensional examination. Sampling of bars, plates, and sheets for visual and dimensional examination shall be in accordance with the provisions of MIL-STD-105, inspection level II, acceptable quality level (AQL) 1.5 percent defective.

4.3.2 For packaging, packing, and marking. A random sample of bundles, lifts, and shipping containers for the inspection of 4.4.2 shall be selected in accordance with the provisions of MIL-STD-105, inspection level II, AQL 4.0 percent defective.

4 3.3 Sampling for tests.

4.3.3.1 For chemical analysis. Two samples shall be taken from each melt in accordance with the procedures of FED-STD-151 for chemical analysis. Samples shall consist of not less than 4 ounces of drillings obtained, if practicable, from a point representing the topmost central portion of the ingot as cast.

4.3.3.2 For magnetic effect. Sampling of bars, plates, and sheets shall comply with the requirements of MIL-M-19595.

4.4 Inspection.

4.4.1 Visual and dimensional examination. Bars, plates, and sheets shall be inspected to determine conformance to the dimensional requirements of 3.4 and the workmanship requirements of 3.5

4.4.2 Packaging, packing, and marking. The packaging and packing of bars, plates, and sheets and marking of packages and containers shall be inspected for conformance to the requirements of section 5.

4.4.3 Tests.

4.4.3.1 Chemical composition. Samples taken in accordance with 4.3.3.1 shall be tested for chemical composition in accordance with the procedures of FED-STD-151 to determine conformance to 3.2. If any sample fails to conform to the requirement of 3.2 the lot represented by the sample shall be rejected.

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4.4.3.2 Magnetic effect. Samples taken in accordance with 4.3.3.2 shall be tested in accordance with the procedures of MIL-M-19595 to determine conformance to the requirements of 3.3.

4.5 Resubmission of rejected lots. Rejected lots may be resubmitted for inspections and tests specified herein in accordance with the provisions of MIL-STD-105.

5. PREPARATION FOR DELIVERY

5.1 General requirements. Bars, plates, and sheets shall be packaged or packed in accordance with one of the following levels as specified in the contract or order. Each package and pack shall contain material of one size and one form and temper.

5.2 Preservation and packaging. Preservation and packaging of the bars, plates, and sheets shall be level A or C, as specified, and in accordance with MIL-P-3993.

5.3 Packing. Packing of the bars, plates, and sheets shall be level A, B, or C, as specified, and in accordance with MIL-P-3993.

5.4 Marking. In addition to any special marking specified in the contract or order, marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The material covered by this specification is intended for special purpose tools and equipment where minimum magnetic effects are required. The limits of magnetic effect are far more stringent than those required for general purpose "nonmagnetic" tools.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Form and temper of material required (see 1.2)
(c) Dimensions and tolerances of bars, plates, and sheets (see 3.4)
(d) Level of packaging or packing required (see 5.1)
(e) Special marking when required (see 5.4).

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6.3 Mechanical properties. The copper-nickel-manganese alloy covered by this specification is capable of developing the mechanical properties shown in table II when heat treated as indicated.

Table II

MECHANICAL PROPERTIES

Heat treatment	Hardness (Rockwell)	Tensile strength (psi)	Yield strength 0.1% offset (psi)	Elongation in 2 inches (%)
Solution treated for 1 hour at 1050° F, water quench	B80	-	-	-
Aged 4 hours at 750° F	C32	124,000	83,000	25.7
Aged 8 hours at 750° F	C40	173,000	135,000	9.1
Aged 16 hours at 750° F	C43	200,000	185,000	4.3
Cold worked and aged 16 hours at 750° F	C48	210,000	-	2.8

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: MIL-C-22226A (OS), Copper-Nickel-Manganese Alloy Bars, Plates, and Sheets, Low Magnetic Effect (For Special Purpose Tools and Equipment)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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